

CS357 Lab 5: Verifying Sequential Code & Loops

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November 05, 2025

In this lab, we use Hoare logic to prove the correctness of programs by using just pens and paper. Show your solutions once you finish all the questions.

1 Example

Here is an example:

```
 $x = 5$       //precondition  
x=x+1;  
 $x = 6$       //postcondition
```

We want to prove its correctness. By the rule Assignment the precondition is calculated as $x = 6[x+1/x]$, which is equivalent to $x+1 = 6$, which is equivalent to $x = 5$. Hence, we write

```
 $x = 5$       //precondition  
 $\Rightarrow$   
 $x + 1 = 6$    //using the 'assignment' rule  
x=x+1;  
 $x = 6$       //postcondition
```

2 Sequential Programs

Prove the following programs by using appropriate Hoare logic rules. Clearly, label where you have used Hoare rules for the assignment and composition in your proofs.

- ```
 $2x = C$ //precondition
t=x;
x=x-y;
y=t+y;
 $x + y = C$ //postcondition
```

2.  $N = qD + r$     *//precondition*  
 $r -= D$ ;  
 $q++$ ;  
 $N = qD + r$     *//postcondition*
3.  $x = A^i$     *//precondition*  
 $x *= A$ ;  
 $i++$ ;  
 $x = A^i$     *//postcondition*
4.  $x = A^{N-i}$     *//precondition*  
 $i--$ ;  
 $x *= A$ ;  
 $x = A^{N-i}$     *//postcondition*
5.  $z = Xi$     *//precondition*  
 $i++$ ;  
 $z += X$ ;  
 $z = Xi$     *//postcondition*
6.  $z = X(Y - i)$     *//precondition*  
 $i--$ ;  
 $z += X$ ;  
 $z = X(Y - i)$     *//postcondition*

### 3 Loop Structures

Prove the partial correctness of the following programs. Remember you need to clearly identify a loop invariant before you can use relevant Hore-logic rules.

1.  $a = x$ ;  
 $y = 1$ ;  
 $\text{while } (a > 0) \{$   
 $\quad y *= a$ ;  
 $\quad a--$ ;  
 $\}$

The precondition here is  $x \geq 0$  and postcondition is  $y = x!$ .

2.  $x = 1$ ;  
 $i = 0$ ;  
 $\text{while } (i \neq N) \{$   
 $\quad x *= A$ ;

```

 i++;
 }

```

The precondition here is  $N \geq 0$  and postcondition is  $x = A^N$ .

3.

```

 i=Y;
 z=0;
 while (i!=0){
 i--;
 z+=X;
 }

```

The precondition here is  $X \geq 0 \wedge Y \geq 0$  and postcondition is  $z = X \cdot Y$ .